

Tawadros Nemer

AI Engineer (Python, PyTorch) | tawadrosgamal@gmail.com | +201098759027 | linkedin.com/in/tawadrosnemer
github.com/TawadrosGamal | Cairo, Egypt (open to relocate)

Summary

AI Engineer with 5+ years of experience in production AI, operations research, and scalable ML systems. Led agentic workflows at Microsoft's Edge Shopping team as primary analytics contact for Copilot Shopping Agent. Expertise spans LLM/RAG systems, forecasting, and full ML lifecycle across Azure, AWS, and GCP. Proven ability to translate ambiguous business problems into production-grade analytical solutions driving measurable impact.

Work Experience

- ✖ **Senior AI Engineer**, CNTXT.AI (Tech) (Remote) Dec 2025 – Present
 - **TESTAI Platform**: Architected and deployed a production platform for testing Speech AI Agents with automated scenario generation, evaluation pipelines, and real-time PII redaction.
 - **Accounting OS AI**: Led development of a multi-service pipeline for email classification, invoice extraction, and line-item classification using GCP (Vertex AI, BigQuery, Pub/Sub) with FastAPI backend.
 - Designed high-throughput, low-latency inference pipelines optimized for production workloads with strict SLAs.
- 🇺🇸 **Applied Scientist**, Microsoft (Tech) (Onsite) May 2024 – Nov 2025
 - Led LLM-powered semantic automation systems in the Edge Shopping team, streamlining product tracking workflows and improving accuracy across data pipelines.
 - Served as the analytics point of contact for Copilot Shopping Agent and Edge Shopping, driving insight generation and data-driven optimization.
 - Orchestrated scalable big-data workflows by automating core processes, URL normalization, CSS-selector maintenance, and comprehensive validation for both implicit and explicit product tracking, resulting in a twofold increase in daily tracked items.
 - Enhanced user engagement by boosting notification CTR and reducing DSATs through data heuristics and behavior modeling.
 - Optimized real-time dashboards and internal tooling (Power BI, custom scripts), reducing latency and improving monitoring.
- 🎓 **Course Author**, Udacity (Education) (Remote) Jun 2024 – Present
 - Authored Course 4: Multi-Agents for the Azure Agentic AI Nanodegree, focusing on practical agentic architectures and applied reasoning.
 - Designed hands-on labs integrating Semantic Kernel and Azure AI Services for multi-agent coordination and workflow automation.
 - Developed examples showcasing autonomous task routing, context memory, and inter-agent collaboration.
- 💻 **Mid-Senior Big Data Engineer**, LigaData (Telecommunication) (Remote) Aug 2023 – May 2024
 - Orchestrated Big Data workflows with Airflow improving efficiency 35% and scalability.
 - Implemented petabyte-scale analytics using Presto/Hive/Elastic Stack.
 - Leveraged advanced SQL for complex data extraction, transformation, and analysis to solve critical business problems in production environments.
 - Built and maintained 120+ automated ETL pipelines using Airflow and Spark; responsible for production monitoring, incident recovery, and maintaining 99.9% pipeline uptime.
 - Mentored junior engineers and migrated 50+ workflows to internal tools.
- 🏠 **Data Scientist & ML Engineer**, Minapharma (HealthCare) (Onsite) Aug 2022 – Aug 2023
 - Engineered Internal sales forecasting solution improving accuracy 120% with Prophet/Neural Networks.
 - Built an ATS tool combining OCR and NLP that improved candidate matching precision by 85%.
 - Developed phone detection model and face recognition model 2x faster than Azure, cutting costs 35%.
- 🏠 **AI Engineer**, CliniDo (HealthCare) (Onsite) Sep 2021 – Aug 2022
 - Engineered NER-driven medical symptom and Arabic sentiment analysis pipelines (88% / 97% accuracy), streamlining diagnostics and elevating patient feedback insights.
 - Integrated Speech-API voice search, reducing query latency by 30% and enhancing user engagement in clinical workflows.
- 🌱 **AI Engineer**, Upwork (Remote) Oct 2020 – May 2022
 - Delivered ETL/EDA/predictive models for 5+ clients across industries.

Education

B.Sc. Computer Science (Very Good with Honors), Assiut University – Assiut, Egypt

Projects

Daleel AI Chat Platform

Architected and deployed an enterprise-grade multimodal AI chat platform on AWS, serving internal knowledge bases and external e-commerce product catalogs. Developed a complex ingestion pipeline that parses unstructured text, extracts tables, and targets embedded graphics utilizing Gemma 4 Vision, indexing assets simultaneously into a vectorized database and a Knowledge Graph. Optimized cost and strict local compliance by migrating baseline inference from Ollama to vLLM, hosting quantized local models (Gemma 4 2B, Qwen 3.5) with Qwen embedding backends. Implemented reliable asynchronous job scheduling for heavy document parsing tasks using RabbitMQ queues.

Logistics Routing Optimization Engine

Built a production-ready vehicle routing problem (VRP) solver using Google OR-Tools. Supports capacity constraints, time windows, and multiple depots. Implemented heuristic search (guided local search) and metaheuristics to minimize total travel distance and fleet size. Integrated real-world distance matrices via OSRM API. Deployed as a FastAPI microservice with Docker. Reduced simulated delivery costs by 22% compared to naive assignment.

Speech-to-Speech AI Call Service for AI Agent Testing

Engineered an end-to-end Speech-to-Speech AI call service that autonomously calls and tests AI voice agents. Implemented using LiveKit for real-time WebRTC communication, integrated OpenAI STS (Speech-to-Speech) models with custom TTS and STT pipelines. The system conducts automated conversations, evaluates responses, and provides comprehensive performance analytics for quality assurance.

Multi-Agent Banking RAG System

Architected a multi-agent RAG system with stateful memory and optimized query routing for financial services.

Real-Time Agentic Customer Service Workflow

Built a real-time agentic workflow using LiveKit and Twilio to automate customer service, cutting handle time by half.

Personalized Recommendation System (SaaS)

Developed RAG-enhanced models increasing relevance 78% and engagement 25% (Python, LLMs).

Medical Diagnosis Assistant

Engineered disease identification system with 95% accuracy using clinical NLP and graph databases.

Multi-Agent SQL Assistant with LLMs

Developed a multi-agent system using LangChain and LangGraph to convert natural language questions into SQL queries, execute them, and return natural language summaries. Implemented embedding-based schema linking with ChromaDB, SQL validation, caching, and audit logging. Supports SQLite and MySQL with robust error handling and retry logic.

RAG-Based Question Generation System

Built a production-ready system that generates high-quality multiple-choice questions from PDF documents using RAG. Leverages Gemini Vision for multi-modal extraction (text, equations, images) to markdown, OpenAI embeddings for semantic search with ChromaDB, and a multi-agent orchestrator (Generator + Evaluator) with LangGraph for quality control. Containerized with Docker and exposed via FastAPI.

Technical Skills

Programming: SQL, Python, R, NoSQL, Scala, C++, Java, C, C#, Git, Pytest, Unit Testing, Problem Solving

Machine Learning: XGBoost, LightGBM, Random Forest, SVM, k-Means, PCA, TensorFlow, PyTorch, Keras, FastAI, ARIMA, Prophet, LSTM, Time Series, Ensemble Learning, Hyperparameter Tuning, Sentiment Analysis

Generative AI: LLMs, Hugging Face (Transformers, Datasets, PEFT, Accelerate), OpenAI, Autogen, LangChain, LangGraph, RAG, Agentic Workflows, Transformers (BERT, GPT, Llama), Gemini, MCP, LlamaIndex, LiveKit, Twilio, Semantic Kernel, Quantization (INT4/INT8/GPTQ), Pydantic, Redis, TTS/STT Systems, **LoRA, QLoRA, PEFT, RLHF, DPO, SFT, vLLM, ONNX / TensorRT**

Speech Technologies: LiveKit, WebRTC, OpenAI STS Models, Speech-to-Speech Pipelines, Voice AI Testing, torchaudio, librosa, soundfile

Data Engineering: Hadoop, Apache Spark, Airflow, ChromaDB, FAISS, Big Data, dbt, DVC, RabbitMQ, Knowledge Graphs

Visualization: Power BI, Tableau, Kibana, Streamlit, Matplotlib, Seaborn, Grafana, Apache Superset

MLOps & Experiment Tracking: Docker, Kubernetes, Flask, FastAPI, TensorFlow Serving, CI/CD, REST APIs, Linux, n8n, Vertex AI, **Weights & Biases, MLflow, DeepSpeed, FSDP, AWS**

NLP & Computer Vision: OCR, Sentiment Analysis, Text Classification, YOLO, SSD, OpenCV, Image Processing

Statistical Modeling: Analytical Problem-Solving, Data Cleaning, Feature Engineering, Hypothesis Testing, Regression, Bayesian Inference

Languages

Arabic (Native), English (Fluent), German (Beginner)